

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 22 Microbiology of the Built Environment

22.1 Multiple Choice Questions

1) Which statement BEST describes sewage and wastewater?

- A) Wastewater is a subset of sewage.
- B) Sewage is a subset of wastewater.
- C) Wastewater and sewage are virtually synonymous.
- D) Wastewater and sewage are both subsets of all liquid effluents from domestic and industrial sources.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

2) Microbes are sometimes used in secondary treatment of wastewater to

- A) convert heavy metals into carbon dioxide.
- B) remove organic matter.
- C) remove large particles.
- D) convert heavy metals into carbon dioxide, remove organic matter, and remove large particles.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

3) With increased levels of oxidizable materials, the biochemical oxygen demand (BOD) will

- A) increase.
- B) decrease.
- C) remain the same.
- D) increase or decrease depending on the nature of the materials involved.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

4) Water leaving an efficient wastewater treatment facility ideally has a biochemical oxygen demand (BOD) between _____ units.

- A) 50 and 100
- B) 25 and 50
- C) 5 and 25
- D) 0 and 5

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

5) Which treatment method(s) are used in primary wastewater treatment?

- A) biological
- B) chemical
- C) physical separation
- D) biological, chemical, and physical separation

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

6) Which product of anoxic sewage treatment can be used to heat and power an entire water treatment facility?

- A) CO₂
- B) H₂
- C) CH₄
- D) O₂

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

7) Aerobic secondary treatment is usually best for _____ wastewater.

- A) industrial
- B) agricultural
- C) residential
- D) chemical

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

8) During secondary aerobic wastewater treatment, some of the activated sludge goes _____ while the rest goes _____.

- A) back to the aerator / to the anoxic sludge digester
- B) back to the primary system / to the aerator
- C) to the anoxic digester / back to the primary system
- D) back to the primary system / to the tertiary digester

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

9) While wastewater is in an activated sludge tank, much of the organic matter is

- A) oxidized.
- B) adsorbed to the floc.
- C) incorporated into microbial cells.
- D) oxidized, adsorbed to the floc, and incorporated into microbial cells.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.6

10) To further reduce the possibility of biological contamination, most treatment plants treat the effluent with

- A) fluorine.
- B) chlorine.
- C) ozone.
- D) extreme heat.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.8

11) What is the advantage of enhanced biological phosphorus removal (EBPR) over chemical precipitation?

- A) Chemical precipitation produces sludge containing arsenic.
- B) EBPR removes significantly more phosphorus than chemical precipitation.
- C) Significantly less sludge is produced by EBPR compared to chemical precipitation.
- D) EBPR is a more stable process.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.7

12) Secondary disinfection of drinking water is necessary to

- A) maintain sufficient residual in the water distribution system to inhibit microbial growth.
- B) prevent the formation of biofilms on pipes in the water distribution system.
- C) neutralize taste- and odor-producing organic compounds.
- D) kill *Cryptosporidium* and other resistant pathogens.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.8

13) Effluent water is wastewater

- A) discharged from the pharmaceutical, plastics, and petrochemical industries.
- B) discharged from the wastewater treatment facility.
- C) that has undergone anoxic secondary wastewater treatment.
- D) that is imported to the wastewater treatment facility for treatment.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

14) Microbes are involved in which step(s) of wastewater treatment?

- A) primary only
- B) secondary and tertiary
- C) secondary only
- D) primary and secondary

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

15) Which of the following bacterial pathogens is found in water distribution systems?

- A) *Enterobacter aerogenes*
- B) *Klebsiella pneumoniae*
- C) *Legionella pneumophila*
- D) *Vibrio cholerae*

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.9

16) The use of _____ to treat drinking water, which began in 1913, reduces the incidence of waterborne diseases.

- A) coagulation
- B) chlorination
- C) filtration
- D) UV irradiation

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.8

17) Which process produces a gaseous product that can be used as fuel to provide heat or power for a wastewater treatment plant?

- A) trickling filter secondary wastewater treatment
- B) aerobic secondary wastewater treatment
- C) anaerobic secondary wastewater treatment
- D) primary wastewater treatment

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.6

18) The goal of wastewater treatment is to

- A) produce potable water.
- B) remove pathogens from wastewater before it is released.
- C) reduce organic and inorganic materials in wastewater to a level that no longer supports microbial growth.
- D) remove pathogens from wastewater to produce potable water.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.8

19) Water distribution systems contain

- A) very few bacteria or other microorganisms.
- B) biofilms that may harbor opportunistic pathogens.
- C) numerous grazing protists that consume bacteria.
- D) biofilms that may harbor opportunistic pathogens and numerous grazing protists that consume bacteria.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.9

20) Opportunistic pathogens such as *Legionella*, *Pseudomonas*, and *Mycobacterium* species have the ability to grow within _____, which may enhance their ability to survive in water distribution systems.

- A) human saliva
- B) fish
- C) protists
- D) UV irradiated water

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 21.9

21) The goal of tertiary wastewater treatment is to sharply reduce the levels of _____ from the final effluent.

- A) organic compounds
- B) heavy metals
- C) methane
- D) inorganic nutrients

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

22) Treatment of drinking water by UV irradiation is effective at _____, but is less effective at _____.

- A) killing bacteria and some cysts / killing viruses
- B) killing viruses / killing bacteria and eukaryotic pathogens
- C) removing odor- and taste-altering organic compounds / killing bacteria
- D) killing bacteria and viruses / killing eukaryotic pathogens

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.8

23) Which of the following is the first, or initiator, reaction in acid mine drainage and leaching?

- A) oxidation of reduced sulfides to sulfate and the release of reduced iron (Fe^{2+})
- B) rapid spontaneous reduction of oxidized iron (Fe^{3+})
- C) rapid oxidation of sulfides to sulfate by oxidized iron (Fe^{3+})
- D) microbial sulfate reduction

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.2

24) What role does *Acidithiobacillus ferrooxidans* play in the copper leaching process?

- A) oxidation of sulfide to sulfate
- B) reduction of oxidized iron (Fe^{3+}) to reduced iron (Fe^{2+})
- C) oxidation of reduced iron (Fe^{2+}) to oxidized iron (Fe^{3+})
- D) reduction of sulfate to sulfide

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.1

25) What chemical initiates the formation of acid mine drainage when pyrite is not disposed of properly?

- A) O_2
- B) Fe^{2+}
- C) HS^-
- D) SO_4^{2-}

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.2

26) The propagation cycle that occurs in acid mine drainage is caused by

- A) reduction of Fe^{3+} by sulfuric acid.
- B) the oxidation of sulfide to sulfuric acid by microbial oxidation of iron (Fe^{3+}).
- C) microbial sulfate reduction.
- D) low pH and high oxygen concentrations.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.2

27) Bioremediation of uranium depends on

- A) fermentation and oxidation.
- B) methanogens.
- C) metal-reducing bacteria.
- D) sulfur-oxidizing bacteria.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.3

28) Microbial immobilization is the best strategy for uranium bioremediation, because

- A) uranium is difficult to chemically oxidize.
- B) uranium contamination is widespread and thus difficult to physically remove.
- C) the by-products of chemical immobilization are toxic.
- D) immobilization may only be temporary.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.3

29) Which of the following can be completely degraded to carbon dioxide by microorganisms?

- A) petroleum
- B) uranium
- C) chlorinated organics
- D) petroleum and chlorinated organics

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.4, 22.5

30) Based on your knowledge of hydrocarbon degradation, what factors most likely limited the rate of petroleum degradation after the Exxon Valdez oil spill off the coast of Alaska?

- A) salt concentration
- B) temperature
- C) inorganic nutrient concentrations
- D) temperature and inorganic nutrient concentrations

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.4

31) Which of the following hydrocarbon compounds are most recalcitrant to degradation during bioremediation, meaning that they stay in the environment longer than the others?

- A) straight-chain hydrocarbons
- B) methane
- C) polycyclic hydrocarbons
- D) natural gases such as ethane and propane

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.4

32) Reductive dechlorination is important in the breakdown of chlorinated xenobiotics because

A) anoxic conditions develop rapidly in polluted habitats and reductive dechlorination can take place in these conditions.

- B) reductive dechlorination breaks down all halogen compounds.
- C) aerobic breakdown of xenobiotics is not possible under any circumstances.
- D) these compounds do not leach into groundwater or sediment.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 21.5

33) Reductive dechlorination is a process in which

- A) oxygenase enzymes are used to add oxygen to chlorinated compounds.
- B) chlorinated organics are used as terminal electron acceptors in anaerobic respiration, releasing chloride.
- C) carbon dioxide is the primary end product.
- D) only dichloro compounds can be degraded.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.5

34) Microbial plastics are composed of

- A) polysaccharides.
- B) Polyhydroxyalkanoates (PHAs).
- C) polyaromatic hydrocarbons.
- D) xenobiotics.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.5

35) Synthetic plastics are

- A) oxidized completely to carbon dioxide after a few years under aerobic conditions.
- B) slower to degrade than microbial plastics.
- C) constructed from PHAs.
- D) more expensive than microbial plastics.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.5

36) To prevent the souring of crude oil, _____ is added to oil well injection water to encourage _____ instead of _____.

- A) oxygen / aerobic respiration / sulfate reduction
- B) sulfate / sulfate reduction / fermentation
- C) sulfate / sulfate reduction / nitrate reduction
- D) nitrate / nitrate reduction / sulfate reduction

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.11

37) Which group of organisms is thought to be responsible for the corrosion of submerged metal structures and pipelines?

- A) fermentative microorganisms
- B) acidophilic iron-oxidizers
- C) sulfate reducers
- D) hydrocarbon-degrading anaerobes

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.11

38) The production of _____ by _____ is thought to be a major contributor to the biodeterioration of stone building materials.

- A) oxalic acid / fungi
- B) sulfuric acid / sulfate reducers
- C) nitric acid / nitrate reducers
- D) carbonic acid / lichens

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.12

39) Crown corrosion of concrete wastewater distribution systems is the result of

- A) acidic fermentation products reacting with the gypsum in the concrete.
- B) sulfate reduction in the sewage and sulfide oxidation in the oxic headspace above.
- C) fungal growth in the concrete.
- D) rapid iron reduction coupled to pyrite dissolution.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.12

40) Why are some xenobiotics biodegraded slower than crude oil and petroleum products?

- A) Xenobiotics were designed to inhibit microbial growth.
- B) Some xenobiotics are novel compounds that differ from naturally occurring chemicals, thus microorganisms have not yet evolved to effectively degrade them.
- C) Only cometabolic microbial processes degrade xenobiotics.
- D) Chlorinated compounds are so toxic that biodegradation is not possible.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.5

22.2 True/False Questions

1) BOTH physical and chemical methods are used to treat and purify drinking water.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.8

2) most wastewater treatment facilities employ methods designed to detect each pathogenic organism that may be present in a given sample.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.6

3) Wastewater plants are usually constructed to handle both domestic and industrial wastes.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

4) Secondary treatment may be aerobic or anoxic.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

5) Potable water is NOT safe for human consumption.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.8

6) In water, most taste- and odor-producing compounds are inorganic in nature.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.8

7) Chlorine is consumed when it reacts with organic materials.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.8

8) Water disinfection agents such as chlorine can easily kill *Cryptosporidium*.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.8

9) It is possible to completely eliminate pathogens from water if sufficient amounts of chlorine are used.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.9

10) Subway air and water distribution systems harbor dangerous levels of pathogenic microorganisms.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.10

11) Pyrite (FeS_2) is common in bituminous coal and metal ores and is a critical component of the microbial leaching of ores and acid mine drainage.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.2

12) Metal-reducing *Geobacter* and *Shewanella* species are important in the microbial mining of metals.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.1, 22.3

13) Inorganic pollutants such as metals and radionuclides can be completely destroyed through bioremediation.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.3

14) Bioremediation of uranium depends on metal-reducing bacteria to change the oxidation state of the uranium to form insoluble uranium compounds.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.3

15) Organisms that degrade hydrocarbons in crude oil are ubiquitous in the environment and have evolved special strategies, such as surfactant production, that allow them to readily use hydrocarbons as electron donors and carbon sources.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.4

16) Chlorinated xenobiotics can ONLY be degraded aerobically, thus making them very recalcitrant if they accumulate in anoxic environments.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.5

17) Although the sulfate-reducing bacteria are thought to be critical in the corrosion of metal structures, the mechanism of metal corrosion is not well understood.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.11

18) Fungi are the main contributor to the biodeterioration of stone building materials and concrete wastewater distribution systems.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 22.12

22.3 Essay Questions

1) What microbial processes are involved in the bioremediation of organic pollutants such as hydrocarbons and the bioremediation of inorganic pollutants such as uranium? What are the end products of bioremediation for hydrocarbons and uranium?

Answer: The bioremediation of hydrocarbons is generally accomplished through respiration of the hydrocarbons to carbon dioxide. This usually occurs in aerobic environments, although slower rates of degradation also occur in the absence of oxygen. In this way, organic pollutants can be almost completely removed from the environment by naturally occurring bacteria. Bioremediation of inorganic pollutants such as uranium does not involve mineralization or complete destruction of the pollutant. In general, anaerobic bacteria are used to change the oxidation state of the metal to a less toxic or insoluble form, thus reducing the impact of the pollutant on the environment. Inorganic pollutants cannot be completely removed by bioremediation.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 22.3, 22.4

2) Natural underground pyritic deposits that are not disturbed by mining activities do not form acidic drainage waters. Why not? Explain the critical steps in the formation of acid mine drainage in your answer.

Answer: The initiation, or first, step in the formation of acid requires the exposure of pyrite to oxygen. If this step does not occur, the propagation cycle that produces sulfuric acid and dissolved metals cannot proceed. When pyrite is exposed to oxygen, it is slowly oxidized into sulfate. The production of sulfate slowly creates an acidic environment and reduced Fe^{2+} is released. Once an acidic environment with Fe^{2+} is established, growth of *Acidithiobacillus ferrooxidans* catalyzes the rapid oxidation of Fe^{2+} to Fe^{3+} under acidic conditions. The Fe^{3+} begins to react rapidly with the pyrite, causing a rapid release of sulfuric acid in the propagation cycle. The acid and any co-occurring metals that are dissolved by the acid then flow away from the exposed pyrite, contaminating the environment.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 22.2

3) Describe the general processes that occur in a sludge digester. What is an advantage of sludge digestion over activated sludge?

Answer: Secondary anaerobic wastewater treatments such as the use of sludge digesters or bioreactors involve fermentation reactions to degrade insoluble organic material like cellulose and fiber. The insoluble matter is first solubilized by anaerobes, and products such as acetate are released. Ultimately, methanogens carry the treatment process to completion by converting acetate into methane and CO₂. An advantage is that the methane produced can be burned as a fuel for heat or energy and less sludge is produced in relation to the amount of organic material degraded.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

4) Compare the operation and function of a trickling filter system and activated sludge.

Answer: Both systems are used as a secondary aerobic treatment method to oxidize organic material. A trickling filter system involves wastewater being slowly passed through a thick layer of crushed rocks. The large surface area of the rock allows for sufficient oxygenation. The rock's surface is coated with a biofilm that actively removes soluble organic material in the wastewater and mineralizes it to CO₂. Activated sludge wastewater is continually mixed and aerated in large tanks. Extensive flocs of microbial biomass form, which absorb and mineralize the organic material to CO₂. After effluent leaves the aeration tank, the flocs are allowed to settle. Some of the floc is returned to the tank to maintain sufficient biomass and a high rate of mineralization. This is unnecessary in a trickling filter system because the biomass is attached to the rock in an extensive biofilm. Both systems use oxygen to mineralize organic matter in wastewater, in contrast to an anaerobic bioreactor.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 22.6

5) Describe floc formation in the activated sludge process.

Answer: During the activated sludge process, continuous aeration and mixing provide sufficient oxygen levels for certain aerobic bacteria (e.g., *Zooglea ramigera*) to produce slime. The slime-forming bacteria enable large aggregates (also called flocs) to form and contain all types of microbes in the activated sludge, such as bacteria, fungi, and protists that can actively transform organic compounds in the wastewater.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 22.6

6) Which microorganisms commonly thrive in drinking water distribution systems and how might they impact human health?

Answer: Despite sufficient chlorination of drinking water, biofilms persist on the walls of pipes in water distribution systems. The biofilms consist mostly of non-pathogenic bacteria, but can harbor opportunistic pathogens such as *Naegleria*, *Acanthamoeba*, *Legionella*, *Pseudomonas*, and *Mycobacterium* species. There are also large numbers of protists present that graze on (eat) the bacteria in the biofilm. *Legionella*, *Pseudomonas*, and *Mycobacterium* species can live inside these protists, increasing their ability to persist in the environment. Although these organisms are present in drinking water systems, they are usually opportunistic pathogens and thus may not affect the health of people with normal immune systems. The impact of these organisms and how they might be distributed through showerheads and other mechanisms is an area of current research.

Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 22.9

7) What is souring of crude oil and how does the petroleum industry control it?

Answer: The souring of crude oil occurs when oil containing sulfur or sulfate from marine injection water becomes contaminated with sulfide through the microbial process of sulfate reduction. The presence of sulfide in the oil causes the degradation of oil pipelines and holding tanks. The petroleum industry controls sulfate reduction by adding nitrate to oil well injection water. Since nitrate respiration is energetically more favorable than sulfate respiration, the nitrate reducers outcompete sulfate reducers in the oil and sulfides are not produced. In fact, nitrate also stimulates the growth of sulfide-oxidizing, nitrate-reducing microorganisms removing sulfides in the crude oil.

Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 22.11

8) If you were in charge of preserving ancient stone ruins and sculptures in an outdoor environment, what strategies would you use to prevent deterioration of the stone?

Answer: Biodeterioration is important to the overall deterioration of stone materials over time. A complex community of microorganisms develops on the surface and inside the stone material. Fungi and fungi-algae symbioses are thought to have a large role in biodeterioration of stone through the production of oxalic acid, which dissolves minerals. Strategies to prevent deterioration could involve protecting the objects from light (to inhibit the growth of lichens) or application of mild anti-fungal solutions. Any strategies related to decreasing fungal growth are appropriate.

Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 22.12